Is MPC Necessary? A Flatness Control for Lateral Mining Vehicle Guidance in Structured Environments

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Abstract: Autonomous mining adoption has not progressed at the same rate as in the conventional automotive industry due to the challenging operating conditions and constraints present in mining environments. These include limited perception caused by dust, mud, and harsh environmental conditions, as well as complex dynamic behavior of heavy-duty vehicles operating under large payloads. To enable wider deployment of intelligent mining systems, motion control algorithms must be tailored to the specific characteristics of these environments. In structured, low-speed scenarios such as mining haul roads, lightweight control strategies may offer a viable alternative to computationally intensive approaches. This paper investigates the use of differential flatness and geometric control for lateral vehicle guidance and compares their performance against a widely used Model Predictive Control (MPC) baseline. The evaluation focuses on trajectory tracking performance, including RMS lateral error, maximum deviation, and heading error, as well as control effort metrics such as steering smoothness and steering rate. Validation is conducted using a mining-relevant high-fidelity simulation environment based on CARLA 0.10.0, incorporating a heavy-duty vehicle model and realistic terrain conditions. Results highlight the operating regimes in which lightweight control strategies can achieve comparable tracking performance to MPC while significantly reducing computational cost and implementation complexity.

Keywords: Autonomous mining, lateral control, differential flatness, geometric control, model predictive control (MPC), trajectory tracking, heavy-duty vehicles, CARLA simulation.

1. INTRODUCTION

Latin America plays a critical role in the global mining industry due to its vast reserves of strategic minerals such as copper, lithium, iron ore, and silver. The region accounts for a significant share of global production, including approximately 41% of copper, 30% of lithium, and nearly 50% of silver output, positioning it as a key supplier for global industrial and energy transitions (GlobalData, 2026). Mining activities represent a major economic driver in several countries across the region. For instance, in Chile, the mining sector contributes between 12% and 14% of national GDP and accounts for more than half of total exports, while in Peru it contributes over 8% of GDP (International Trade Administration, 2025; Hernandez and Perez, 2024). Additionally, the sector plays a crucial role in employment generation, providing hundreds of thousands of direct jobs and supporting a broader ecosystem of indirect employment through services, logistics, and infrastructure development (Statista, 2025).

Beyond its economic significance, the mining sector is undergoing a rapid technological transformation driven by automation, digitalization, and the integration of advanced analytics and artificial intelligence. Recent reports indicate that approximately 31% of mining companies in Latin America are actively adopting AI-driven solutions to improve efficiency, safety, and operational performance (Bujakiewicz, 2025). These advancements are enabling the development of smart mining systems, where data-driven decision-making and automation play a central role in optimizing production and reducing operational risks (Digital Market Latin America, 2024). Despite these advancements, the adoption of autonomous systems in mining remains slower than in the conventional automotive sector due to challenging environmental conditions, including reduced visibility caused by dust and mud, as well as the complex dynamics of heavy-duty vehicles operating under extreme loads. These constraints highlight the need for robust and efficient motion control strategies specifically tailored to mining environments.

Autonomous mining systems have gained increasing attention in recent decades due to the need for improving safety, operational efficiency, and productivity in hazardous environments. Mining environments are inherently challenging, characterized by confined spaces, poor visibility conditions, uneven terrain, and the presence of heavy-duty vehicles operating under extreme payloads. These factors make manual operation both risky and inefficient, motivating the adoption of intelligent and unmanned mining technologies GlobalData (2026). In particular, autonomous haulage systems and unmanned equipment have been successfully deployed in both surface and underground mines, where centralized control, wireless communication, and real-time monitoring enable continuous operation with reduced human intervention Li and Zhan (2018).

From a technological perspective, autonomous mining vehicles rely on a hierarchical architecture that integrates perception, motion planning, and trajectory tracking control Chen et al. (2023). Among these components, trajectory tracking—especially lateral control—plays a critical role in ensuring that heavy-duty vehicles follow predefined paths accurately and safely. However, the characteristics of mining vehicles differ significantly from conventional passenger vehicles. Mining dump trucks are typically large-scale systems with masses exceeding hundreds of tonnes, subject to variable payloads, actuator delays, and uncertain dynamics, which complicates both modeling and control design Chen et al. (2023).

Several control strategies have been proposed for autonomous vehicle path tracking, including geometric methods such as pure pursuit and Stanley control, as well as model-based approaches like Model Predictive Control (MPC). While MPC has demonstrated superior performance compared to classical methods in many scenarios, particularly due to its ability to handle constraints and optimize future control actions, it also introduces significant computational complexity Lalezar et al. (2024). This trade-off becomes particularly relevant in mining applications, where real-time implementation and system robustness are critical.

In addition to control, path planning plays a fundamental role in mining vehicle operation. Smooth and obstacle-avoiding trajectories are essential to ensure safe navigation and efficient operation. Previous studies have shown that the use of curvature-optimized B-spline paths can significantly improve traversal speed and reduce mechanical stress on articulated mining vehicles, highlighting the importance of path geometry in overall system performance Berglund et al. (2010). Furthermore, smoother trajectories reduce steering effort and enable higher operational speeds without compromising safety, which is particularly important for heavy articulated vehicles in constrained environments.

Despite these advancements, the majority of existing research has focused either on high-performance control methods such as MPC or on improvements in planning and perception. Comparatively fewer studies have explored lightweight control strategies tailored to the specific conditions of mining environments, particularly those characterized by low-speed, structured trajectories such as haul roads. This gap motivates the exploration of alternative control approaches that can achieve comparable tracking performance with reduced computational requirements and implementation complexity.

2. LATERAL DYNAMIC MODEL OF A MINING DUMP TRUCK

While the kinematic bicycle model is commonly used in the automotive domain for passenger vehicles, its assumptions are no longer sufficient for heavy-duty mining vehicles. Due to the significantly increased forces associated with large payloads, a dynamic bicycle model is required to accurately capture vehicle behavior. In particular, effects such as tire side-slip, oversteering, and friction forces become dominant and must be explicitly considered.

Dynamic models provide a more accurate representation of the forces acting on the vehicle, especially tire forces, which play a critical role in lateral stability and trajectory tracking. Although a full four-wheel dynamic model can capture these effects with high fidelity, it results in a highly nonlinear and complex system that is difficult to use for control design Lee et al. (2008).

Since full vehicle models are typically used for validation and high-fidelity simulation purposes, their complexity makes them less suitable for real-time controller development. As a result, simplified dynamic models—such as the two-wheel (bicycle) dynamic model—are commonly adopted in control applications Kebbati et al. (2023). These models provide a balance between accuracy and computational efficiency.

The dynamic bicycle model considers planar motion of the vehicle, including longitudinal motion along the x-axis and lateral motion along the y-axis. Considering the bicycle model illustrated in Fig. 1, and applying Newton's laws of motion, the vehicle dynamics can be described by the following equations:

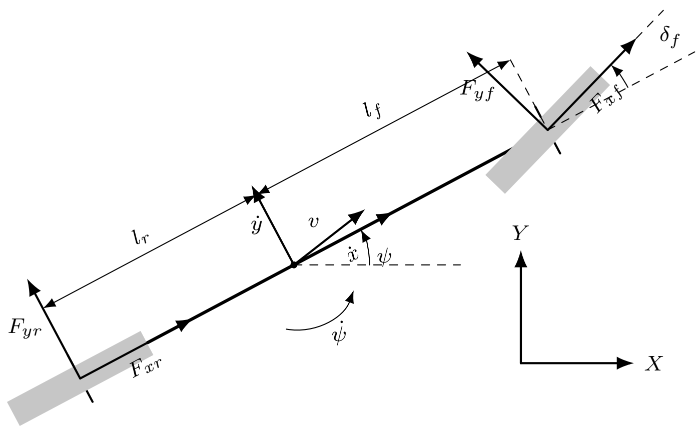

Fig. 1. Bicycle model representation for a mining dump truck.

$$\ddot{y} = \left(\frac{C_r l_r - C_f l_f}{m\dot{x}} - \dot{x}\right)\dot{\psi} - \left(\frac{C_f + C_r}{m\dot{x}}\right)\dot{y} + \frac{C_f}{m}\,\delta_f \quad (1)$$

$$\ddot{\psi} = -\left(\frac{C_f l_f^2 + C_r l_r^2}{I_z\dot{x}}\right)\dot{\psi} + \left(\frac{C_r l_r - C_f l_f}{I_z\dot{x}}\right)\dot{y} + \frac{C_f l_f}{I_z}\,\delta_f \quad (2)$$

$$\dot{Y} = \dot{x}\sin\psi + \dot{y}\cos\psi \approx \dot{x}\psi + \dot{y} \quad (3)$$

The lateral tire forces are modeled using a linear tire model, where the lateral force is proportional to the slip angle:

$$F_{yf} = C_f \alpha_f, \qquad F_{yr} = C_r \alpha_r \tag{4}$$

Under the small-angle assumption, the slip angles can be approximated as:

$$\alpha_f \approx \delta_f - \frac{\dot{y} + l_f \dot{\psi}}{\dot{x}}, \qquad \alpha_r \approx -\frac{\dot{y} - l_r \dot{\psi}}{\dot{x}} \tag{5}$$

The lateral dynamic model is described by a set of states associated with the planar motion of the vehicle. In particular, the system states are the lateral velocity $\dot{y}$, the yaw rate $\dot{\psi}$, the yaw angle ψ, and the global lateral displacement Y. The longitudinal velocity $\dot{x}$ is treated as a measurable scheduling variable and is assumed to remain constant or slowly varying over the prediction horizon. The steering angle at the front axle, δ_f, is considered the control input. The physical parameters of the model are given by the vehicle mass m, the yaw moment of inertia I_z, the distances l_f and l_r from the center of gravity to the front and rear axles, and the tire cornering stiffness coefficients C_f and C_r. The lateral tire forces at the front and rear axles are denoted by F_{yf} and F_{yr}, respectively, while F_{xf} and F_{xr} represent the longitudinal tire forces. The tire slip angles, α_f and α_r, are introduced to model the lateral tire behavior through a linear tire approximation valid under small slip-angle conditions.

Table 1. Definition of variables and parameters used in the lateral dynamic model.

Symbol	Definition
$\dot{x}$	Longitudinal velocity of the vehicle
$\dot{y}$	Lateral velocity at the center of gravity (CG)
$\ddot{y}$	Lateral acceleration
ψ	Yaw angle (vehicle heading angle)
$\dot{\psi}$	Yaw rate
$\ddot{\psi}$	Yaw acceleration
Y	Global lateral position
$\dot{Y}$	Global lateral velocity
δ_f	Front steering angle (control input)
m	Total vehicle mass
I_z	Yaw moment of inertia about the vertical axis
l_f	Distance from the CG to the front axle
l_r	Distance from the CG to the rear axle
C_f	Front tire cornering stiffness
C_r	Rear tire cornering stiffness
F_{yf}	Front lateral tire force
F_{yr}	Rear lateral tire force
F_{xf}	Front longitudinal tire force
F_{xr}	Rear longitudinal tire force
α_f	Front tire slip angle
α_r	Rear tire slip angle
v	Resultant vehicle speed at the CG

2.1 Differential Flatness-Based Reference Generation

The planar motion of the vehicle can be described using a kinematic bicycle model, which is known to be differentially flat. A nonlinear system is said to be differentially flat if there exists an output vector z such that all system states and inputs can be expressed as functions of z and a finite number of its derivatives.

For the planar vehicle model, the flat output is selected as

$$z = \begin{bmatrix} X \\ Y \end{bmatrix}. \tag{6}$$

All relevant states and inputs can be reconstructed from the flat output as follows:

2.1.0.1. Heading angle

$$\psi = \operatorname{atan2}(\dot{Y}, \dot{X}) \tag{7}$$

2.1.0.2. Longitudinal velocity

$$v = \sqrt{\dot{X}^2 + \dot{Y}^2} \tag{8}$$

2.1.0.3. Path curvature

$$\kappa = \frac{\dot{X}\ddot{Y} - \dot{Y}\ddot{X}}{(\dot{X}^2 + \dot{Y}^2)^{3/2}} \tag{9}$$

2.1.0.4. Steering angle

$$\delta_{ff} = \arctan(L\kappa) \tag{10}$$

This formulation enables direct generation of dynamically feasible trajectories and the corresponding feedforward steering input from a desired path.

2.2 Reference Trajectory Generation: Lane Change

To evaluate the proposed controller, a smooth lane-change maneuver is selected as the reference trajectory. The longitudinal motion is defined as

$$X_d(t) = v_x t \tag{11}$$

The lateral displacement is defined using a normalized cubic polynomial that ensures smooth transitions:

$$Y_d(t) = y_f \left(3\tau^2 - 2\tau^3\right), \quad \tau = \frac{t}{T} \tag{12}$$

where y_f is the desired lateral displacement and T is the maneuver duration.

The first and second derivatives are given by:

$$\dot{Y}_d(t) = y_f \left(\frac{6\tau}{T} - \frac{6\tau^2}{T}\right) \tag{13}$$

$$\ddot{Y}_d(t) = y_f \left(\frac{6}{T^2} - \frac{12\tau}{T^2}\right) \tag{14}$$

The desired heading, yaw rate, and yaw acceleration are computed directly from the flat outputs:

$$\psi_d = \operatorname{atan2}(\dot{Y}_d, \dot{X}_d) \tag{15}$$

$$\dot{\psi}_d = \frac{\dot{X}_d \ddot{Y}_d - \dot{Y}_d \ddot{X}_d}{\dot{X}_d^2 + \dot{Y}_d^2} \tag{16}$$

The corresponding feedforward steering command is obtained as:

$$\delta_{ff} = \arctan(L\kappa_d) \tag{17}$$

2.3 Flatness-Based Tracking Control with Integral Action

To account for model uncertainties and achieve accurate trajectory tracking, a feedback linearization controller augmented with lateral error integral action is proposed.

The yaw dynamics can be expressed as:

$$\ddot{\psi} = A\dot{\psi} + B\dot{y} + C\delta_f \tag{18}$$

Using the kinematic approximation:

$$\dot{y} \approx \dot{Y} - v_x\psi \tag{19}$$

the system is reformulated explicitly in terms of measurable states.

2.3.0.1. Error definitions

$$e_\psi = \psi - \psi_d \tag{20}$$

$$e_{\dot{\psi}} = \dot{\psi} - \dot{\psi}_d \tag{21}$$

$$e_y = -\sin(\psi_d)(X - X_d) + \cos(\psi_d)(Y - Y_d) \tag{22}$$

To compensate for steady-state errors caused by model mismatch, an integral term is introduced:

$$e_y^{int} = \int e_y(t)\,dt \tag{23}$$

2.3.0.2. Control law The final steering input is defined as:

$$\boxed{\begin{aligned}\delta_f = \delta_{ff} + \frac{1}{C}\Big[\ddot{\psi}_d - k_1 e_{\dot{\psi}} - k_0 e_\psi - k_y e_y \\ - k_i e_y^{int} - A\dot{\psi} - B(\dot{Y} - v_x\psi)\Big]\end{aligned}} \tag{24}$$

2.3.0.3. Control interpretation The proposed controller combines three key components:

- Feedforward term δ_{ff} derived from differential flatness, capturing path curvature.
- Feedback linearization terms that cancel vehicle dynamics and enforce desired error dynamics.
- Integral action on lateral error to eliminate steady-state offsets due to modeling inaccuracies.

This structure enables accurate trajectory tracking with reduced computational complexity compared to optimization-based controllers such as MPC.

2.4 Active Disturbance Rejection Control

Mining vehicles operate under highly variable conditions: load changes, uneven terrain, and uncertainty in tire-soil friction. ADRC addresses this by treating all unknown dynamics as a *total disturbance*, estimated and canceled in real time via an Extended State Observer (ESO), without requiring an accurate system model.

Extended State Observer The ESO estimates the global lateral position, its derivative, and the total disturbance:

$$z_1 = \hat{Y}, \quad z_2 = \dot{\hat{Y}}, \quad z_3 = \hat{f} \tag{25}$$

The discrete-time observer equations are:

$$e = Y_m - z_1 \tag{26}$$

$$\begin{aligned}\dot{z}_1 &= z_2 + \beta_1 e \\ \dot{z}_2 &= z_3 + \beta_2 e + b_0\delta_f \\ \dot{z}_3 &= \beta_3 e\end{aligned} \tag{27}$$

where each state is integrated with sampling period T_s:

$$z_i(t + T_s) = z_i(t) + T_s\,\dot{z}_i \tag{28}$$

The observer gains are obtained via bandwidth parameterization:

$$\beta_1 = 3\omega_o, \quad \beta_2 = 3\omega_o^2, \quad \beta_3 = \omega_o^3 \tag{29}$$

2.5 Control Law

The control signal is computed as:

$$u_0 = k_p(Y_d - z_1) + k_d(0 - z_2) \tag{30}$$

$$\boxed{\delta_f = \frac{u_0 - z_3}{b_0}} \tag{31}$$

with $b_0 = C_f/m$, $k_p = \omega_c^2$, $k_d = 2\omega_c$, and $\omega_c \le \omega_o/2$.

3. SIMULATION RESULTS

The performance of the proposed flatness-based controller is evaluated using a lane-change maneuver with a target lateral displacement of $y_f = 3.5$ m.

3.1 Tracking Performance

Fig. 2 shows the lateral displacement tracking results. The controller achieves accurate trajectory tracking with zero steady-state error, demonstrating successful convergence to the desired path. The transient response is smooth and stable, with no oscillatory behavior.

An initial rapid increase in lateral displacement is observed, driven by the feedforward component obtained from trajectory curvature. This is followed by a slower convergence phase associated with the feedback and integral actions, which compensate for model mismatch and dynamic uncertainties.

3.2 Role of Control Components

The results highlight the contribution of each element of the proposed control structure:

- The feedforward term derived from differential flatness provides the baseline steering action required to follow the path geometry.
- Feedback linearization compensates for nonlinear vehicle dynamics and ensures stable transient behavior.
- Integral action eliminates steady-state lateral error caused by modeling inaccuracies and approximations.

Without integral action, the controller exhibits a persistent steady-state tracking error despite achieving correct heading alignment. The inclusion of the integral term enables complete error rejection.

3.3 Discussion

The simulation results demonstrate that the proposed flatness-based controller can achieve high-quality trajectory tracking for heavy-duty vehicles. The controller maintains stability and smoothness while accurately following the desired path.

These results indicate that flatness-based control, combined with feedback linearization and integral correction, provides an effective and computationally efficient alternative to optimization-based approaches for structured, low-speed mining scenarios.

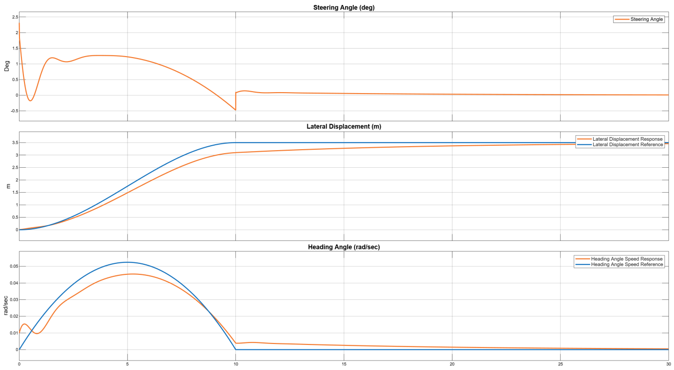

Fig. 2. Lateral displacement tracking performance for the lane-change maneuver. The proposed controller achieves accurate tracking with zero steady-state error.

The simulation results show that the ADRC controller achieves accurate trajectory tracking without requiring an exact dynamic model. The Extended State Observer estimates and compensates for the total disturbance, which encompasses model uncertainties, terrain irregularities, and payload variations. A slight response delay is observed due to the observer convergence time, an inherent characteristic of the ESO that can be mitigated by increasing the observer bandwidth ω_o at the expense of higher noise sensitivity. Despite this limitation, the controller maintains stability and achieves zero steady-state error, offering a robust and computationally lightweight alternative to optimization-based methods.

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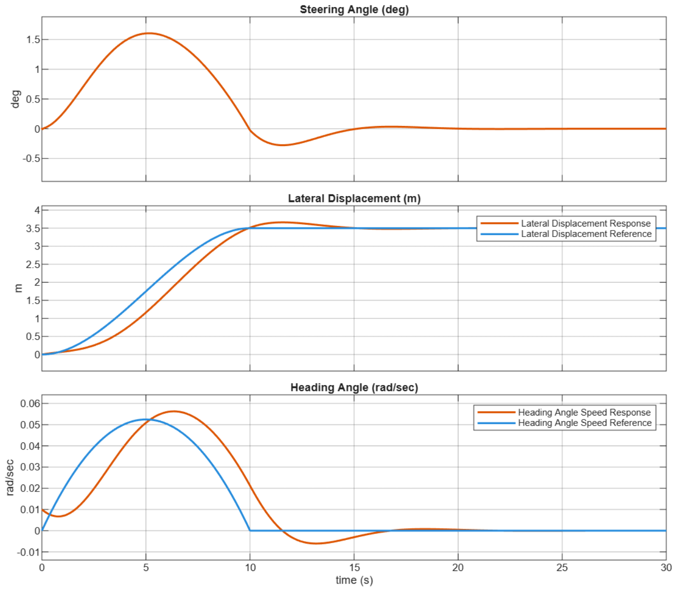

Fig. 3. Lateral displacement tracking performance for the lane-change maneuver using the ADRC-based controller. The ADRC achieves accurate tracking with zero steady-state error despite model uncertainties and external disturbances.

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Appendix A. CAT 797 SPECIFICATION

Using the standard bicycle model approximations, the distances from the center of gravity to the front and rear axles, as well as the yaw moment of inertia, are obtained as:

$$l_f = L\left(1 - \frac{m_f}{m}\right), \quad l_r = L\left(1 - \frac{m_r}{m}\right), \quad I_z = l_f\, l_r\, m \tag{A.1}$$

For the Caterpillar 797F dump truck, the following parameters are used:

$$L = 7.195 \text{ m} \tag{A.2}$$

$$\frac{m_f}{m} = 0.333, \quad \frac{m_r}{m} = 0.667 \tag{A.3}$$

$$m = 623{,}690 \text{ kg} \tag{A.4}$$

Substituting into the equations: s

$$l_f = 7.195 \times (1 - 0.333) = 4.80 \text{ m} \tag{A.5}$$

$$l_r = 7.195 \times (1 - 0.667) = 2.40 \text{ m} \tag{A.6}$$

$$I_z = 4.80 \times 2.40 \times 623{,}690 \approx 7.19 \times 10^6 \text{ kg} \cdot \text{m}^2 \tag{A.7}$$

A.1 Cornering Stiffness Parameters

Unlike geometric parameters, the cornering stiffness coefficients cannot be directly derived from manufacturer data. Therefore, they are estimated using a data-driven least squares method. Additionally, simulation-based estimates obtained from the CARLA environment are used to capture terrain-dependent variations, providing a more realistic representation of tire-road interaction in mining environments.